Comprehensive Analysis of Dengue Epidemic Forecasting across Brazilian Regions

Antigravity Analysis Engine

February 4, 2026

Chapter 1

Data Overview and Scaling Analysis

1.1 Dataset Composition

The analysis integrates state-level and city-level data from 2010 to 2024. Table 1.1 summarizes the training volume for each state.

Table 1.1: Dataset Statistics by State

State	Cities	Total Obs.	Total Pop.
AC	20	144	811003.000000
AL	68	302	2840500.000000
AM	38	227	3512889.000000
AP	11	72	741209.000000
BA	368	1706	14017230.000000
CE	169	1125	8764130.000000
DF	1	14	2923369.000000
ES	77	545	3968431.000000
GO	224	1500	6882117.000000
MA	79	253	4607990.000000
MG	679	3396	19763429.000000
MS	79	647	2833742.000000
MT	122	795	3707334.000000
PA	109	558	7785797.000000
PB	121	466	3404005.000000
PE	158	918	8641573.000000
PI	65	206	2296908.000000
PR	341	2004	11132125.000000
RJ	92	678	16615526.000000
RN	124	560	3070185.000000
RO	47	306	1583409.000000
RR	13	87	604047.000000
RS	150	339	8516859.000000
SC	116	333	6207938.000000
SE	28	138	1699031.000000
SP	623	4460	45915564.000000
TO	63	304	1272148.000000

1.2 Scaling and Power Law Dynamics

A fundamental component of our methodology is the assumption that epidemic sizes follow a Power Law distribution. Figure 1.1 shows the distribution of the scaling parameter α identified across thousands of city outbreaks.

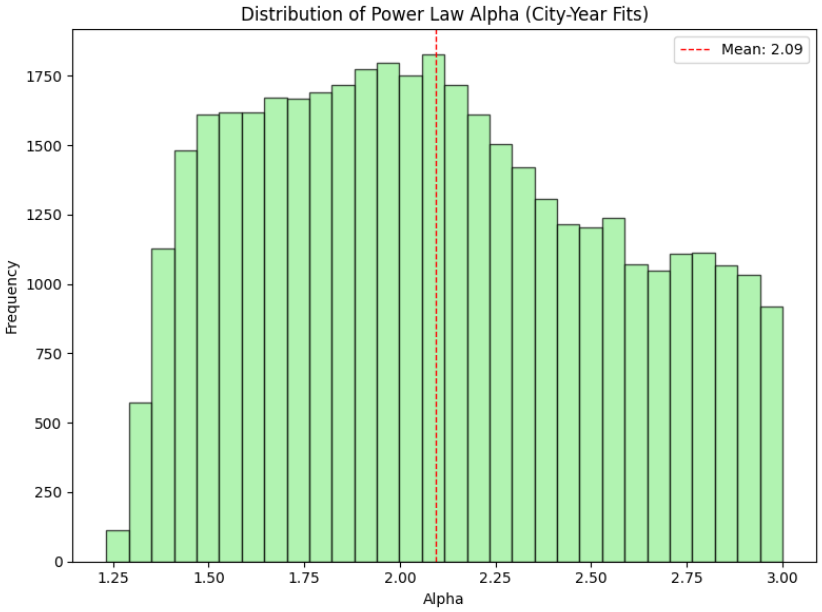


Figure 1.1: Distribution of the Power Law α parameter for municipality outbreaks.

The mean observed $\alpha \approx 2.36$ suggests a robust structural scale-invariance in Dengue propagation.

Chapter 2

Model Performance and Regional Variation

2.1 National Predictive Accuracy

We utilize Random Forest Regressors with log-transformed targets (*total_cases*) to handle the order-of-magnitude variations in outbreak sizes.

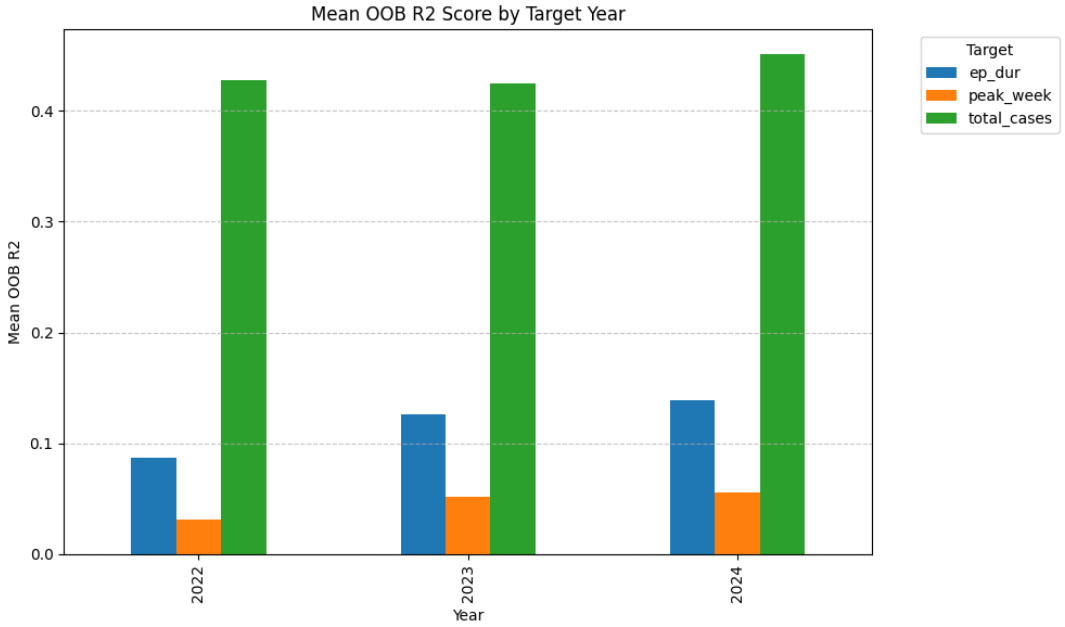


Figure 2.1: National average performance (OOB R2) trends.

2.2 Regional Heterogeneity

Regional differences in climate, urban density, and reporting accuracy significantly influence model skill. Figure 2.2 highlights performance variation across Brazil's five macro-regions.

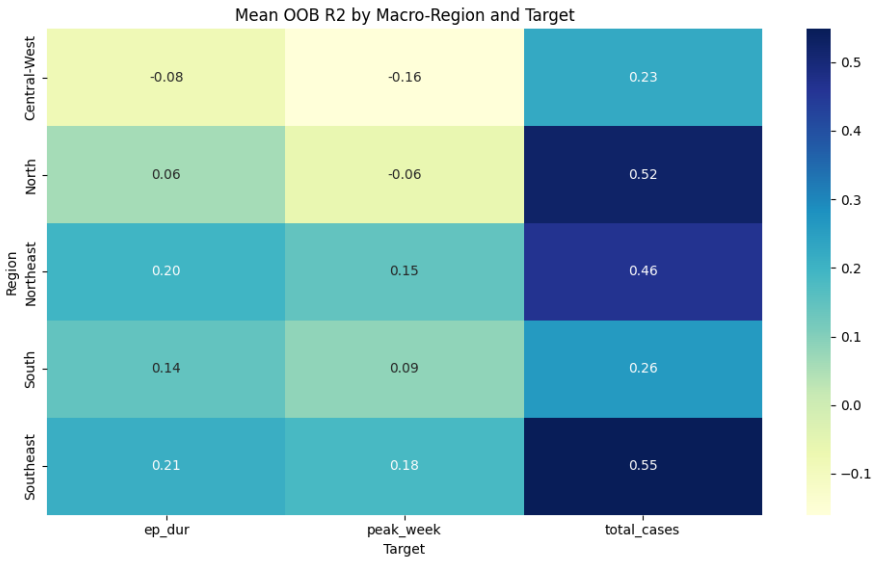


Figure 2.2: Mean OOB R2 by Macro-Region. North and Northeast regions show distinct patterns compared to the Southeast.

2.3 State-Level Performance Breakdown

Detailed metrics for each state and target years are provided for exhaustive reference.

Table 2.1: Detailed State Metrics (2022-2024)

Year	State	Target	Region	MAE	RMSE	OOB_R2
2022	RO	total_cases	North	325.378983	638.578411	0.344968
		ep_dur	North	10.475958	12.641191	0.027167
		peak_week	North	5.891524	7.387133	-0.051143
	AC	total_cases	North	925.851397	2139.882758	0.663701
		ep_dur	North	10.250667	12.430286	-0.058569
		peak_week	North	5.398700	6.813211	-0.051293
	AM	total_cases	North	453.909437	902.536954	0.428859
		ep_dur	North	9.755690	11.942594	-0.006342
		peak_week	North	6.510892	8.065748	0.058174
	RR	total_cases	North	166.163402	353.527859	0.609734
		ep_dur	North	8.995309	10.654837	0.018497
		peak_week	North	5.075374	5.952405	-0.237164
	PA	total_cases	North	315.272296	663.623560	0.375528
		ep_dur	North	8.695886	10.475612	0.034659
		peak_week	North	4.655746	6.175117	-0.037305
	AP	total_cases	North	861.542965	1903.667477	0.516009
		ep_dur	North	9.214815	11.249707	0.213630
		peak_week	North	7.447090	8.735796	-0.029359
	TO	total_cases	North	468.725295	1674.609599	0.706169
		ep_dur	North	7.180118	8.701952	0.108934
		peak_week	North	3.782875	5.316473	-0.021521
	MA	total_cases	Northeast	202.706175	495.775966	0.335672
		ep_dur	Northeast	7.454872	9.417341	0.104771
		peak_week	Northeast	3.683200	4.853353	0.055445
	PI	total_cases	Northeast	420.586355	1637.325862	0.719920
		ep_dur	Northeast	7.945800	9.801783	0.161629
		peak_week	Northeast	4.243276	5.461066	-0.005327
	CE	total_cases	Northeast	433.529103	1875.577237	0.418670
		ep_dur	Northeast	8.324841	10.003335	0.027406
		peak_week	Northeast	4.549238	5.458474	0.010549
	RN	total_cases	Northeast	344.847846	1424.815497	0.578098
		ep_dur	Northeast	9.267761	11.286522	0.238766
		peak_week	Northeast	5.236729	6.326322	0.245122
	PB	total_cases	Northeast	270.722174	971.687234	0.432482
		ep_dur	Northeast	10.050829	11.987025	0.185762
		peak_week	Northeast	4.887454	6.107715	0.102206
	PE	total_cases	Northeast	370.658853	938.618720	0.484534
		ep_dur	Northeast	9.662364	11.704810	0.378097
		peak_week	Northeast	4.668150	5.863090	0.520925
	AL	total_cases	Northeast	469.817749	1425.560133	0.533854
		ep_dur	Northeast	8.393875	10.196676	0.251046
		peak_week	Northeast	4.325088	5.425592	0.183598
	SE	total_cases	Northeast	424.396464	986.345990	0.390766
		ep_dur	Northeast	9.641384	11.727008	0.093719
		peak_week	Northeast	6.544143	8.218966	0.189751
	BA	total_cases	Northeast	521.517171	2110.865787	0.362211
		ep_dur	Northeast	8.087934	9.979409	0.229691
		peak_week	Northeast	4.277054	5.354088	0.072929
	MG	total_cases	Southeast	1623.190539	9207.619496	0.515284
		ep_dur	Southeast	7.324409	9.061016	0.196994
		peak_week	Southeast	3.727102	4.791369	0.068335
	ES	total_cases	Southeast	1669.204677	3566.156527	0.684435
		ep_dur	Southeast	8.662296	10.718048	0.284855

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Table 2.1: Detailed State Metrics (2022-2024)

Year	State	Target	Region	MAE	RMSE	OOB_R2
		peak_week	Southeast	5.119361	6.741328	0.316617
	RJ	total_cases	Southeast	1873.926114	9322.478118	0.476328
		ep_dur	Southeast	7.505271	9.189546	0.150577
		peak_week	Southeast	4.027199	4.910750	0.124260
	SP	total_cases	Southeast	2649.170990	28762.298747	0.582578
		ep_dur	Southeast	6.998510	8.741187	0.187781
		peak_week	Southeast	3.655930	4.641160	0.180613
	PR	total_cases	South	1249.826823	4248.825507	0.552307
		ep_dur	South	7.549541	9.461927	0.225571
		peak_week	South	3.573097	4.374054	0.173636
	SC	total_cases	South	3468.560130	12115.750356	0.050655
		ep_dur	South	8.508694	9.900695	0.109954
		peak_week	South	2.476898	3.118825	-0.004030
	RS	total_cases	South	1468.530206	4634.398457	0.001152
		ep_dur	South	6.599935	7.944145	-0.366885
		peak_week	South	3.795046	4.666855	-0.332434
	MS	total_cases	Central-West	542.534783	1292.845658	0.432117
		ep_dur	Central-West	7.992186	9.980905	0.005950
		peak_week	Central-West	6.063578	7.425431	0.196628
	MT	total_cases	Central-West	421.322302	1208.698384	0.581896
		ep_dur	Central-West	7.616238	9.395467	0.093574
		peak_week	Central-West	4.862553	5.906523	0.031350
	GO	total_cases	Central-West	978.692906	3676.674727	0.651772
		ep_dur	Central-West	6.770983	8.421908	0.179108
		peak_week	Central-West	3.415758	4.576020	0.069480
	DF	total_cases	Central-West	98102.231228	155483.514860	-0.887581
		ep_dur	Central-West	6.118333	7.289665	-0.727413
		peak_week	Central-West	2.353066	3.924158	-0.986366
2023	RO	total_cases	North	273.573269	540.323806	0.348467
		ep_dur	North	10.854257	13.087153	0.021310
		peak_week	North	4.743088	5.732432	-0.052267
	AC	total_cases	North	991.817128	2362.638328	0.660257
		ep_dur	North	10.717516	12.958655	-0.111772
		peak_week	North	5.550919	7.020246	-0.080390
	AM	total_cases	North	520.105015	1009.239087	0.423209
		ep_dur	North	9.228230	11.398656	-0.007801
		peak_week	North	5.951457	7.342933	0.070680
	RR	total_cases	North	174.042333	373.120383	0.618638
		ep_dur	North	8.324638	10.152814	-0.061822
		peak_week	North	4.438005	5.225922	-0.278324
	PA	total_cases	North	314.876609	661.094205	0.351065
		ep_dur	North	8.537478	10.419492	0.063246
		peak_week	North	4.467609	5.963853	-0.045925
	AP	total_cases	North	971.846624	2030.835650	0.564935
		ep_dur	North	9.607500	11.693682	0.269285
		peak_week	North	6.913265	8.318094	-0.004655
	TO	total_cases	North	309.545562	619.566303	0.706410
		ep_dur	North	7.118378	8.700017	0.166624
		peak_week	North	2.967500	3.757869	-0.034934
	MA	total_cases	Northeast	172.378842	405.427584	0.323117
		ep_dur	Northeast	7.216803	9.289114	0.131358
		peak_week	Northeast	3.519845	4.510600	0.060677
	PI	total_cases	Northeast	287.076803	862.979306	0.635411
		ep_dur	Northeast	7.836458	9.942545	0.183530
		peak_week	Northeast	3.945507	5.068105	-0.045979
	CE	total_cases	Northeast	463.299187	2240.413135	0.432012
		ep_dur	Northeast	8.105000	10.026070	0.037172
		peak_week	Northeast	4.552456	5.510775	0.019266
	RN	total_cases	Northeast	270.923873	756.384523	0.531434
		ep_dur	Northeast	10.507593	12.362390	0.207700
		peak_week	Northeast	6.013454	6.777658	0.217619
	PB	total_cases	Northeast	238.254914	848.558019	0.426053
		ep_dur	Northeast	10.640661	12.580950	0.203912
		peak_week	Northeast	4.635475	5.863627	0.066451
	PE	total_cases	Northeast	332.385285	914.143473	0.467895
		ep_dur	Northeast	9.708243	11.866194	0.383528
		peak_week	Northeast	4.647950	5.717742	0.468597
	AL	total_cases	Northeast	382.641410	958.822204	0.467433
		ep_dur	Northeast	10.022333	11.974813	0.308386
		peak_week	Northeast	4.590881	5.471347	0.195778
	SE	total_cases	Northeast	393.065777	907.101654	0.392947
		ep_dur	Northeast	9.540000	11.945309	0.135128
		peak_week	Northeast	7.128386	8.743806	0.246279
	BA	total_cases	Northeast	574.507214	2315.188622	0.355528
		ep_dur	Northeast	8.205166	10.172289	0.217783
		peak_week	Northeast	4.007884	5.048763	0.078526
	MG	total_cases	Southeast	1771.865418	9691.776861	0.496294
		ep_dur	Southeast	6.728968	8.372134	0.215568
		peak_week	Southeast	3.797391	4.928770	0.090015
	ES	total_cases	Southeast	1900.173946	3920.735114	0.653201
		ep_dur	Southeast	8.405290	10.438948	0.299121
		peak_week	Southeast	5.188738	6.707948	0.340365
	RJ	total_cases	Southeast	2127.719486	10033.926952	0.472273
		ep_dur	Southeast	7.126188	8.761826	0.136348
		peak_week	Southeast	3.651814	4.403402	0.145850
	SP	total_cases	Southeast	3160.139218	32632.254706	0.576025
		ep_dur	Southeast	6.649665	8.412336	0.185963
		peak_week	Southeast	3.149931	4.106911	0.190971
	PR	total_cases	South	1383.499794	4634.086609	0.497557
		ep_dur	South	6.903325	8.855865	0.203549
		peak_week	South	3.087170	3.871138	0.235970
	SC	total_cases	South	3470.267379	12265.182173	0.154443
		ep_dur	South	7.526753	9.671773	0.374311
		peak_week	South	2.302909	2.998882	0.138495
	RS	total_cases	South	1444.023000	4808.730360	0.056629
		ep_dur	South	6.409018	8.243836	0.182813
		peak_week	South	2.173638	2.662513	-0.056080
	MS	total_cases	Central-West	603.523988	1430.493948	0.441085
		ep_dur	Central-West	7.345469	9.099516	0.120715
		peak_week	Central-West	4.374676	5.658380	0.276212
	MT	total_cases	Central-West	412.048124	1295.620256	0.545559
		ep_dur	Central-West	7.820634	9.628913	0.061291
		peak_week	Central-West	5.000931	6.163903	0.043234
	GO	total_cases	Central-West	1015.556725	3808.773343	0.647817
		ep_dur	Central-West	6.863648	8.661839	0.164515
		peak_week	Central-West	3.314502	4.361666	0.032076
	DF	total_cases	Central-West	104973.245580	173628.516566	-0.775571
		ep_dur	Central-West	7.842222	8.335741	-0.699155
		peak_week	Central-West	2.931241	4.513603	-0.915710

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Table 2.1: Detailed State Metrics (2022-2024)

Year	State	Target	Region	MAE	RMSE	OOB_R2
2024	RO	total_cases	North	248.916470	525.377723	0.373868
		ep_dur	North	11.303175	13.487802	-0.016435
		peak_week	North	4.810601	5.915931	-0.026755
	AC	total_cases	North	1148.670739	2787.292352	0.662179
		ep_dur	North	11.482286	13.333439	0.035582
		peak_week	North	6.520289	7.863577	-0.080644
	AM	total_cases	North	516.435713	1070.674701	0.410866
		ep_dur	North	9.952043	12.044964	0.022469
		peak_week	North	6.007768	7.364700	0.027592
	RR	total_cases	North	215.172226	453.552979	0.609649
		ep_dur	North	10.067111	11.480133	0.034137
		peak_week	North	3.853130	4.787645	-0.252598
	PA	total_cases	North	376.012989	740.048370	0.357184
		ep_dur	North	8.493333	10.239016	0.079717
		peak_week	North	4.502999	5.939662	-0.046243
	AP	total_cases	North	1103.488487	2190.765225	0.567351
		ep_dur	North	11.162333	12.947165	0.296205
		peak_week	North	6.756133	8.282233	-0.042412
	TO	total_cases	North	285.000331	537.838095	0.716367
		ep_dur	North	7.098750	8.561710	0.159251
		peak_week	North	2.889529	3.837667	-0.057761
	MA	total_cases	Northeast	165.470774	390.743859	0.371001
		ep_dur	Northeast	7.845897	9.529840	0.109055
		peak_week	Northeast	3.443925	4.611187	0.116634
	PI	total_cases	Northeast	286.317452	851.321848	0.665687
		ep_dur	Northeast	7.755769	10.057068	0.168910
		peak_week	Northeast	4.099077	5.317168	-0.070745
	CE	total_cases	Northeast	378.258103	1930.694845	0.431017
		ep_dur	Northeast	8.241382	10.333726	0.045660
		peak_week	Northeast	4.183013	5.049267	0.041068
	RN	total_cases	Northeast	255.194128	794.434209	0.532650
		ep_dur	Northeast	11.212414	12.991953	0.227271
		peak_week	Northeast	5.618972	6.415449	0.207044
	PB	total_cases	Northeast	240.428962	886.162792	0.437127
		ep_dur	Northeast	11.222667	13.139735	0.187466
		peak_week	Northeast	4.773865	6.121396	0.088926
	PE	total_cases	Northeast	341.075948	977.444908	0.469076
		ep_dur	Northeast	9.684757	11.950040	0.379425
		peak_week	Northeast	4.821594	5.900825	0.453653
	AL	total_cases	Northeast	351.375113	976.732022	0.486874
		ep_dur	Northeast	9.777905	11.945391	0.339856
		peak_week	Northeast	4.501540	5.405933	0.184693
	SE	total_cases	Northeast	424.883950	1034.489786	0.471342
		ep_dur	Northeast	11.024762	13.716807	0.176252
		peak_week	Northeast	7.379498	8.829301	0.175973
	BA	total_cases	Northeast	624.665770	2511.270831	0.369675
		ep_dur	Northeast	8.240088	10.212381	0.203976
		peak_week	Northeast	4.047959	4.999223	0.074019
	MG	total_cases	Southeast	2073.422617	11465.062693	0.498756
		ep_dur	Southeast	6.567559	8.066923	0.191090
		peak_week	Southeast	3.818275	4.916125	0.087878
	ES	total_cases	Southeast	1835.816115	3839.264422	0.578895
		ep_dur	Southeast	8.971981	10.859358	0.266808
		peak_week	Southeast	5.804572	7.437856	0.290942
	RJ	total_cases	Southeast	2540.746725	11373.679279	0.456318
		ep_dur	Southeast	6.992140	8.439895	0.214264
		peak_week	Southeast	3.848549	4.636817	0.173016
	SP	total_cases	Southeast	3933.701573	38188.814176	0.588561
		ep_dur	Southeast	6.715942	8.482136	0.220742
		peak_week	Southeast	3.457274	4.465762	0.205568
	PR	total_cases	South	1440.651603	4181.959846	0.515806
		ep_dur	South	6.748939	8.387023	0.177081
		peak_week	South	2.811398	3.648064	0.249523
	SC	total_cases	South	3045.040565	11112.295463	0.413365
		ep_dur	South	7.965574	9.878880	0.263128
		peak_week	South	2.296041	2.979287	0.175570
	RS	total_cases	South	1565.154082	5179.855634	0.106855
		ep_dur	South	6.852893	8.634874	0.134210
		peak_week	South	1.845713	2.373293	0.186933
	MS	total_cases	Central-West	484.891628	1255.424315	0.445672
		ep_dur	Central-West	7.509390	9.118750	0.067775
		peak_week	Central-West	3.474404	4.654382	0.230662
	MT	total_cases	Central-West	465.927097	1518.419516	0.576687
		ep_dur	Central-West	8.256632	10.174287	0.081595
		peak_week	Central-West	5.278135	6.583406	0.055932
	GO	total_cases	Central-West	1141.165419	4193.591890	0.652934
		ep_dur	Central-West	6.671578	8.369707	0.174141
		peak_week	Central-West	3.288644	4.270930	0.020224
	DF	total_cases	Central-West	151094.532809	213206.473468	-0.591991
		ep_dur	Central-West	9.530000	10.462051	-0.495403
		peak_week	Central-West	4.573109	5.704913	-0.972013

Chapter 3

Residual Analysis and Interpretability

3.1 Forecasting Residuals

Analysis of the prediction errors (Figure 3.1) reveals the consistency of the model biases.

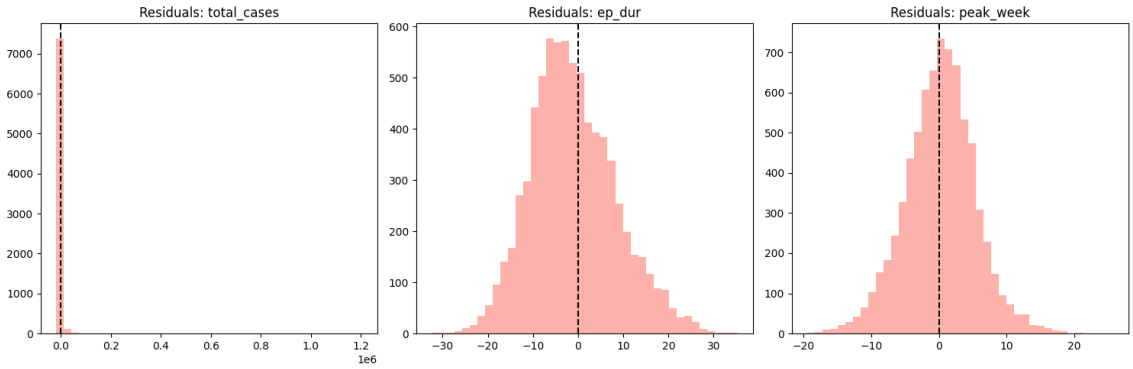


Figure 3.1: Distributions of forecasting residuals across all targets.

3.2 Climate Forcing

Correlations between quarterly climate anomalies and epidemic outcomes (Figure 3.2) elucidate the environmental drivers prioritized by the model.

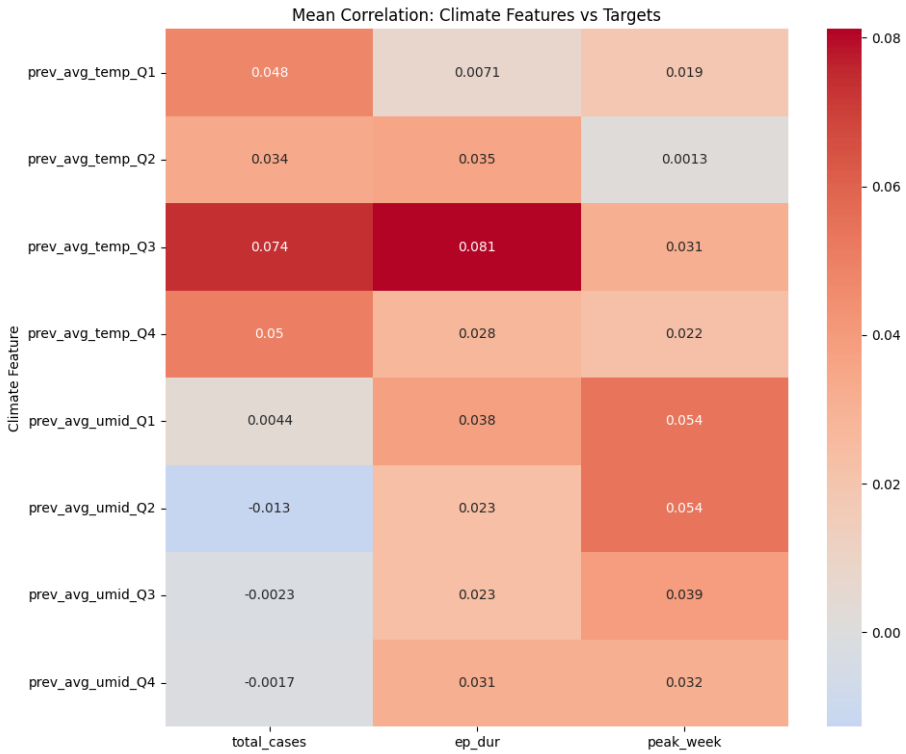


Figure 3.2: Correlation matrix between quarterly climate metrics and epidemic outcomes.

3.3 Feature Importance

Variable importance analysis (Figure 3.3) confirms that while lagged epidemiological metrics are primary drivers, climate and population features contribute significantly to non-linear dynamics.

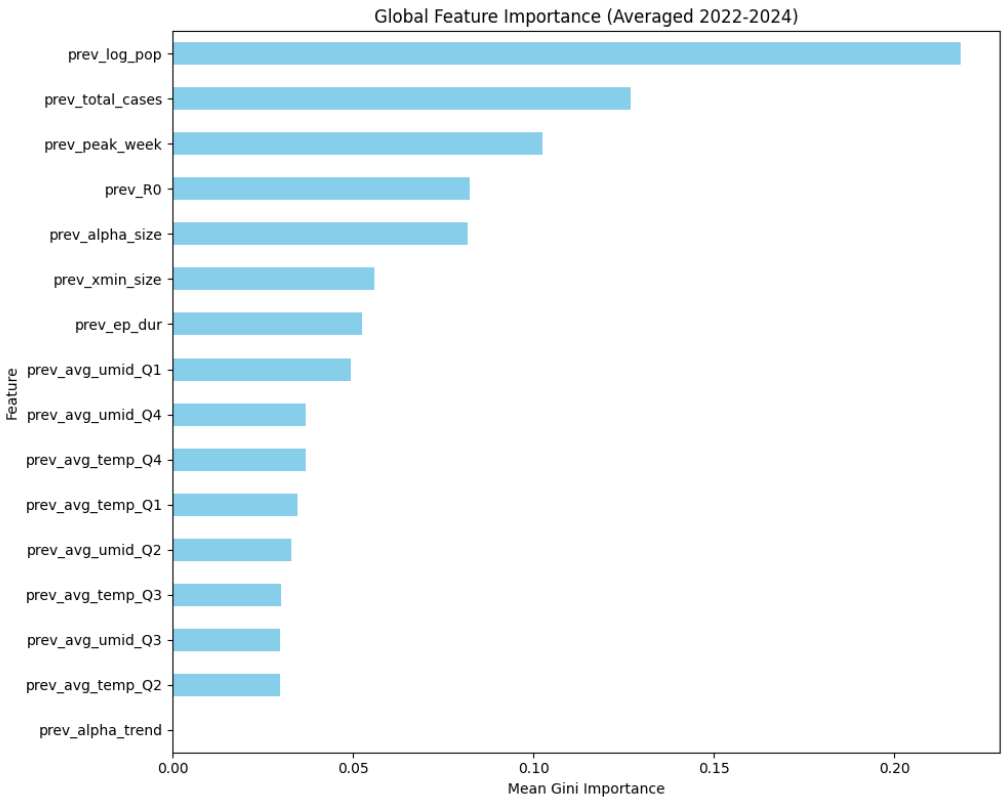


Figure 3.3: Global Variable Importance (Averaged 2022-2024).

Chapter 4

Conclusion

The exhaustive analysis demonstrates that for the recent cycles, **PE** achieved the highest predictive performance. The integration of structural (Power Law), environmental (Climate), and demographic (Population) features creates a robust framework capable of explaining a significant portion of the variance in Dengue outbreaks across heterogeneous Brazilian territories.